

Multiple Choice (10 marks)

1. Are Python variable names case-sensitive?
 - (a) Yes
 - (b) No
 - (c) It's machine-dependent
 - (d) None of the above
2. What is the value of this python expression: $1 + 3 \% 3$?
 - (a) 0
 - (b) 1
 - (c) 3
 - (d) 4
3. In Python 3, what is the output of `print tuple[0]` if `tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )`?
 - (a) ('abcd', 786 , 2.23, 'john', 70.2)
 - (b) abcd
 - (c) Error
 - (d) None of the above.
4. Which of the following best explains what happens when a new device is connected to the Internet?
 - (a) A device driver is assigned to the device.
 - (b) An Internet Protocol (IP) address is assigned to the device.
 - (c) A packet number is assigned to the device.
 - (d) A Web site is assigned to the device.
5. What is the output of `4*1**4` in python?
 - (a) 4
 - (b) 1
 - (c) 256
 - (d) 16

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the `random()` function sets the integer starting value used in generating random numbers.
7. True or False: The condition `a < c` is sufficient to ensure that the expression evaluates to true.
8. True or False: The function `max(l)` in Python 3 returns the second largest element in the list `l`, which is 3 for the list `[1, 2, 3, 4]`.
9. True or False: In Python 3, the `strip([chars])` function removes all leading and trailing whitespace from a string.
10. True or False: In Python 3, the expression `x << 3` results in 3 when `x` is equal to 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of a Web server in the delivery of Web pages, including its functions, how it interacts with clients, and its importance in the overall web ecosystem.
12. Discuss the selection sort algorithm, focusing on how the number of comparisons required to sort a list of numbers is affected by their initial arrangement. Provide examples to support your analysis.

End of Paper